Loan Approval Prediction

Machine Learning for Financial Decision-Making

Made with GAMMA

Project Overview

Data Analysis

Financial and demographic data

ML Modeling

Decision Tree algorithm

Visualization

Insights and predictions

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Dataset Features

Financial Factors

- Income
- Loan Amount
- Credit Score
- Asset Values

Demographic Factors

- Age
- Education
- Employment Status

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Technology Stack

Python

Core programming language

Pandas

Data handling and manipulation

Scikit-learn

Machine learning framework

Matplotlib

Data visualization

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Project Workflow

Data Loading

CSV import

Data Cleaning

Missing values

Feature Engineering

Label encoding

Model Building

Decision Tree

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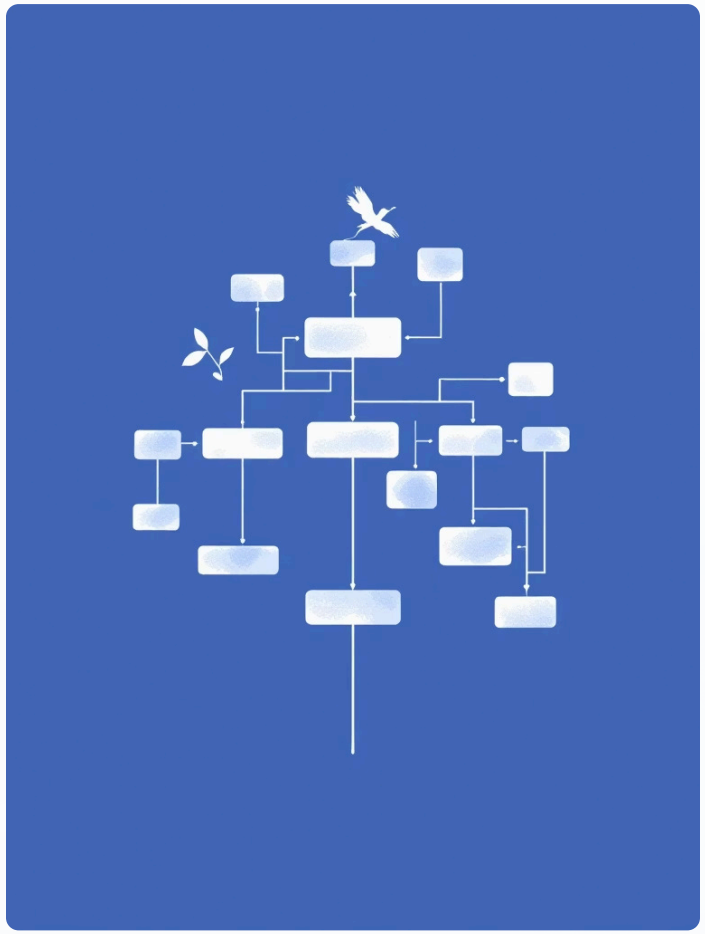
Model Architecture

Decision Tree Classifier

- Entropy-based splitting
- Max depth: 3 levels
- Interpretable rules
- Transparent logic



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Key Insights

Credit Score Impact

Major factor in approval decisions

Income Influence

Direct correlation with approval

Asset Values

Significantly affects outcomes

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Sample Results

3

Approved

Person 1, 3, 4

1

Rejected

Person 2

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Future Improvements

01

Model Evaluation

Accuracy metrics, confusion matrix

02

Advanced Models

Random Forest, XGBoost

03

Hyperparameter Tuning

Optimize model performance

04

Deployment

Streamlit web interface

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Project Structure

Dashboard & Reporting

- Power BI integration
- Approval trends
- Income analysis
- Credit score insights

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